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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

I Semester B. E. Examinations Nov/Dec-18

Common to All Branches

ELEMENTS OF MECHANICAL ENGINEERING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
3. Use of steam tables is permitted.

PART-A

1	1.1	The first law of thermodynamics is called the law of conservation of _____.	01
	1.2	The thermodynamic properties which depend on the mass of the system are called _____ properties	01
	1.3	Calculate the enthalpy of steam at a pressure of 30 bar, if its dryness fraction is 0.75.	02
	1.4	What is the use of the nozzle with needle in Pelton Wheel?	01
	1.5	In Francis turbine, the centre of the runner is fitted with _____ tube, to enable the discharge of water at lower pressure.	01
	1.6	Enlist the Advantages of Open Cycle Gas Turbine.	02
	1.7	During the power stroke of 4-Stroke petrol engine, the power is transmitted to the crankshaft through _____.	01
	1.8	The power developed by an engine at the output shaft is called _____.	01
	1.9	Draw the $P-V$ diagram that represents theoretical Otto cycle.	02
	1.10	_____ is a machine tool which holds the work in the holding device and rotating it against a suitable cutting tool to remove the excess metal.	01
	1.11	_____ is the lathe operation of reducing the diameter of work piece over a narrow surface.	01
	1.12	Sketch straddle milling operation.	02
	1.13	What is welding?	01
	1.14	In soldering, what is the filler metal called as?	01
	1.15	Mention the basic elements of CNC system.	02

PART-B

2	a	State Zeroth law of thermodynamics. What is the basic concept of this law?	04
	b	Explain the following three types steams: i) Dry steam ii) Wet steam iii) Super-heated steam.	06
	c	Calculate the specific volume and enthalpy of $8kg$ of steam at $1.2MPa$ i) When the steam is 12% wet ii) When the steam is superheated at $300^{\circ}C$, $C_P = 2.25KJ/\,Kg\ K$.	06

3	a	With a sketch, explain open cycle gas turbine	04
	b	Give the complete classification of water turbine with an example.	06
	c	With a neat sketch briefly explain the working of a vapour compression refrigeration.	06
		OR	
4	a	Mention two functions of compressor and combustion chamber of gas turbine.	04
	b	With a neat sketch explain the working of Pelton Wheel	06
	c	What are the properties of a good refrigerant? Explain.	06
5	a	Define the following $I.C.$ Engine Terminology: i) Bore ii) Stroke iii) Top Dead Centre(TDC).	04
	b	What are the advantages of two stroke engine over a four stroke engine?	06
	c	A four cylinder, four stroke internal combustion engine develops an indicated power of $50KW$ at $3000rpm$. The cylinder diameter is $75mm$ and the stroke is $90mm$. Find the mean effective pressure in each cylinder. If the mechanical efficiency is 80%. What effective brake load would be required if the effective brake drum diameter is $0.6m$?	06
		OR	
6	a	With the help of a PV diagram explain the working of a four stroke diesel engine.	08
	b	The following observations were recorded during a test on a 4 stroke engine. Bore =$250mm$, stroke =$400mm$,crankspeed =$250rpm$, Net load on the break drum =$700N$. Diameter of the brake drum =$2m$, Indicated mean effective pressure =6bar, Fuel consumption=$0.0013kg/s$,Specific gravity of fuel =0.78, Calorific value of fuel =$43900kJ/kg$. Determine: i) Break power ii) Indicated power iii) Friction power iv) Mechanical efficiency v) Indicated thermal efficiency and vi) Break thermal efficiency.	08
7	a	Explain how the size of a lathe is specified.	04
	b	Differentiate between following drilling operations: i) Drilling and boring ii) Counter sinking and counter boring.	06
	c	List and briefly explain any two operations that can be performed on a milling machine.	06
8	a	Differentiate between arc welding and gas welding	04
	b	With a neat sketch explain oxy-acetylene welding process.	06
	c	What are the advantages and disadvantages of CNC systems over conventional machines?	06